The $2a + 1$ Composition Identity: Uniqueness and Domain Constraint

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Abstract

Building on the structural polynomial framework of [2], we prove two results about the Steiner composition identity $\Phi_{\mathrm{St}(1,\beta)}(\Phi_{\mathrm{St}(\alpha,1)}(a)) = \Phi_{\mathrm{St}(\alpha+1,\beta+2)}(2a+1)$.

First, a uniqueness theorem: this identity holds for all $\alpha \geq 1$, $\beta \geq 0$, and all admissible a if and only if $g = 3$, $h = 2$, and $q = 1$, uniquely characterising the standard Collatz basis among all generalised (g, h, q) maps with $g, h \geq 2$ and $g \neq h$. The proof works entirely in $\mathbb{Z}[g, h]$: the coefficient-of-a equation forces $h = 2$; comparing the constant-term equation at $\alpha = 1$ and $\alpha = 2$ forces $g = 3$; and substituting $(g, h) = (3, 2)$ then forces $q = 1$.

Second, a domain constraint: in a concrete Collatz orbit, the exponent β is not a free parameter but is determined by a via $\beta = v_2(3^{\alpha+1}m - 1) - 2$, where $a + 1 = 2^\alpha m$ with m odd. The case $\beta = 1$ is characterised by $c \equiv 1 \pmod 8$ (equivalently $3^\alpha m \equiv 3 \pmod{16}$), where $c = \Phi_{\mathrm{St}(\alpha,1)}(a)$ is the intermediate value.

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1 Introduction

Let (g, h, q) be a generalised Collatz-type basis, with odd step $x \mapsto gx + q$ and even step $x \mapsto x/h$. The Steiner word $\mathrm{St}(\alpha, \beta)$ denotes the parity sequence consisting of α repetitions of OE followed by β additional E-steps; at the standard Collatz basis $(3, 2, 1)$ this encodes α Syracuse steps with $\alpha + \beta$ total halvings.

In [2] we developed the structural polynomial $k_w(g, h) \in \mathbb{Z}[g, h]$ attached to a parity sequence w, and established the path identity $h^e \cdot b = g^o \cdot a + q \cdot k_w(g, h)$, the geometric polynomial theorem $k_{\mathrm{St}(\alpha,\beta)}(g, h) = (g^\alpha - h^\alpha)/(g - h)$, a composition rule, and a cycle element identity.

This paper applies those tools to investigate the identity

$$\Phi_{\mathrm{St}(1,\beta)}\big(\Phi_{\mathrm{St}(\alpha,1)}(a)\big) \;=\; \Phi_{\mathrm{St}(\alpha+1,\,\beta+2)}(2a + 1), \tag{1}$$

which holds at the Collatz basis and relates siblings $\{a, 2a+1\}$ in the Collatz tree. Section 2 proves that (1) characterises the Collatz basis uniquely. Section 3 derives the β-formula and its modular characterisation.

2 The $2a + 1$ Identity and Its Uniqueness

We use the notation of [2] throughout. In particular, $\Phi_w(a)$ denotes the affine map corresponding to the parity sequence w, and $k_w(g, h)$ is the structural polynomial defined there.

Definition 2.1 (The $2a + 1$ composition identity)**.** We say that the basis (g, h, q) *satisfies the* $2a + 1$ *identity* for parameters $\alpha \geq 1$, $\beta \geq 0$ if (1) holds for all a in the domain of definition.

Theorem 2.2 (Uniqueness of the $2a + 1$ identity)**.** *Let (g, h, q) be a generalised Collatz-type basis with $g, h \geq 2$ and $g \neq h$. The identity* (1) *holds for all $\alpha \geq 1$, $\beta \geq 0$, and all admissible a if and only if $g = 3$, $h = 2$, and $q = 1$.*

Proof. We compute both sides as affine maps in a, working in $\mathbb{Z}[g, h]$, using the path identity, the geometric polynomial theorem, and the composition rule from [2].

Step 1: the left-hand side. Apply the composition rule with $P = \mathrm{St}(\alpha, 1)$ and $Q = \mathrm{St}(1, \beta)$. We have $o_P = \alpha$, $e_P = \alpha + 1$, $o_Q = 1$, $e_Q = 1 + \beta$. From the geometric polynomial theorem:

$$k_P(g, h) = \frac{g^\alpha - h^\alpha}{g - h}, \qquad k_Q(g, h) = \frac{g - h}{g - h} = 1.$$

The composition rule gives

$$k_{Q\circ P}(g, h) = g \cdot \frac{g^\alpha - h^\alpha}{g - h} + h^{\alpha+1}.$$

Comparing with $k_{\mathrm{St}(\alpha+1,\cdot)} = (g^{\alpha+1} - h^{\alpha+1})/(g - h)$, the difference is

$$k_{Q\circ P} - \frac{g^{\alpha+1} - h^{\alpha+1}}{g - h} = \frac{h^\alpha(h - 1)(g - h)}{g - h} = h^\alpha(h - 1),$$

so

$$k_{Q\circ P}(g,h) = \frac{g^{\alpha+1} - h^{\alpha+1}}{g-h} + h^{\alpha}(h-1). \tag{2}$$

The LHS map therefore satisfies

$$h^{\alpha+\beta+2} \cdot \Phi_{Q\circ P}(a) = g^{\alpha+1}a + q\left(\frac{g^{\alpha+1} - h^{\alpha+1}}{g-h} + h^{\alpha}(h-1)\right). \tag{3}$$

Step 2: the right-hand side. $\mathrm{St}(\alpha+1, \beta+2)$ has $o = \alpha+1$, $e = \alpha+\beta+3$, structural polynomial $(g^{\alpha+1} - h^{\alpha+1})/(g-h)$. Applied to $2a+1$:

$$h^{\alpha+\beta+3} \cdot \Phi_{\mathrm{St}(\alpha+1,\beta+2)}(2a+1) = g^{\alpha+1}(2a+1) + q \cdot \frac{g^{\alpha+1} - h^{\alpha+1}}{g-h}. \tag{4}$$

Step 3: equating. For (1) to hold, multiply (3) by h and equate with (4). Equating coefficients of a:

$$hg^{\alpha+1} = 2g^{\alpha+1} \quad \Longrightarrow \quad h = 2.$$

Equating constant terms and substituting $h = 2$:

$$q\left(\frac{g^{\alpha+1} - 2^{\alpha+1}}{g-2} + 2^{\alpha+1}\right) = g^{\alpha+1}. \tag{5}$$

For this to hold for *all* $\alpha \geq 1$, we evaluate at $\alpha = 1$ and $\alpha = 2$ (with $h = 2$ substituted throughout):

$$\begin{aligned}
\alpha = 1: &\quad q\left(\frac{g^2-4}{g-2} + 4\right) = q(g+2+4) = q(g+6) = g^2, \\
\alpha = 2: &\quad q\left(\frac{g^3-8}{g-2} + 8\right) = q(g^2+2g+4+8) = q(g^2+2g+12) = g^3.
\end{aligned}$$

Both instances fix q uniquely:

$$q = \frac{g^2}{g+6}, \qquad q = \frac{g^3}{g^2+2g+12}.$$

Setting these equal and clearing denominators:

$$g^2(g^2+2g+12) = g^3(g+6) \implies 12g^2 = 4g^3 \implies g = 3.$$

(Since $g \geq 2$, we may divide by g^2.)

Substituting $g = 3$ into (5): $q \cdot 3^{\alpha+1} = 3^{\alpha+1}$ for all $\alpha \geq 1$, hence $q = 1$.

With $g = 3$, $h = 2$, $q = 1$: the linear coefficients agree since $h = 2$. For the constants, equation (5) gives $q \cdot 3^{\alpha+1} = 3^{\alpha+1}$, which holds for $q = 1$. Therefore the affine maps agree for all $\alpha \geq 1$, $\beta \geq 0$, and all admissible a, confirming sufficiency. □

Corollary 2.3 (Collatz uniqueness)**.** *Among all generalised Collatz-type bases (g, h, q) with $g, h \geq 2$ and $g \neq h$, the identity (1) holds (for all α, β) if and only if $(g, h, q) = (3, 2, 1)$.*

Proof. Theorem 2.2 shows the three constraints $h = 2$, $g = 3$, and $q = 1$ are each necessary and together sufficient. □

Remark 2.4. The algebraic structure of the proof makes the three constraints transparent. The condition $h = 2$ comes from matching the linear coefficient of a: the even step must be division by 2. The condition $g = 3$ comes from requiring the constant-term equation to hold for *all* α: $3x + 1$ is the unique odd step compatible with the Steiner sibling structure. The condition $q = 1$ then follows immediately by substituting $g = 3$, $h = 2$ into the constant-term equation. Working in $\mathbb{Z}[g, h]$ separates all three constraints clearly, in a way that numerical verification at $(g, h, q) = (3, 2, 1)$ cannot.

3 The Domain Constraint and Modular Characterisation

The identity (1) is an algebraic statement about affine maps that holds for any fixed $\beta \geq 0$. In a concrete Collatz orbit, β is *determined* by a: it is the number of trailing even steps after the first odd step following $c = \Phi_{\mathrm{St}(\alpha,1)}(a)$. This section makes that determination explicit.

We write $b := \Phi_{\mathrm{St}(1,\beta)}(c) = \Phi_{\mathrm{St}(\alpha+1,\beta+2)}(2a + 1)$ for the endpoint and $c := \Phi_{\mathrm{St}(\alpha,1)}(a)$ for the intermediate value.

3.1 Domain of $\mathrm{St}(\alpha, 1)$

Write $a + 1 = 2^\alpha m$ with m odd.

Lemma 3.1 (Intermediate value formula)**.** *Let $a + 1 = 2^\alpha m$ with m odd. Then*

$$c \;=\; \Phi_{\mathrm{St}(\alpha,1)}(a) \;=\; \frac{3^\alpha m - 1}{2}.$$

Moreover, c is odd if and only if $3^\alpha m \equiv 3 \pmod 4$.

Proof. By the geometric polynomial theorem [2], $k_{\mathrm{St}(\alpha,1)}(3, 2) = 3^\alpha - 2^\alpha$. The affine map is $c = (3^\alpha a + 3^\alpha - 2^\alpha)/2^{\alpha+1}$. Substituting $a = 2^\alpha m - 1$:

$$c = \frac{3^\alpha \cdot 2^\alpha m - 2^\alpha}{2^{\alpha+1}} = \frac{3^\alpha m - 1}{2}.$$

Since $3^\alpha m$ is odd, c is an integer. It is odd iff $3^\alpha m \equiv 3 \pmod 4$. □

3.2 The β-formula

Definition 3.2 (Trailing halving count β)**.** When c is odd, the O-step sends $c \mapsto 3c + 1$, which is even. Exactly $v_2(3c+1)$ E-steps are then required to reach the next odd state. The next Steiner word is $\mathrm{St}(1, \beta) = \mathtt{OE}\,\mathtt{E}^\beta$: one E-step is already the $\mathtt{E}$ in $\mathtt{OE}$, and the remaining β E-steps are the trailing $\mathtt{E}^\beta$. Hence $1 + \beta = v_2(3c + 1)$, giving

$$\beta \;:=\; v_2(3c + 1) - 1,$$

where v_2 denotes the 2-adic valuation. (The case $\beta = 0$ corresponds to $v_2(3c + 1) = 1$, i.e. $3c + 1 \equiv 2 \pmod 4$.)

Lemma 3.3 (The β-formula)**.** *With $a+1 = 2^\alpha m$ (m odd) and $c = (3^\alpha m - 1)/2$ odd,*

$$\beta \;=\; v_2\big(3^{\alpha+1}m - 1\big) - 2. \tag{6}$$

Proof. Direct computation:

$$3c + 1 = \frac{3(3^\alpha m - 1)}{2} + 1 = \frac{3^{\alpha+1}m - 1}{2}.$$

So $v_2(3c+1) = v_2(3^{\alpha+1}m - 1) - 1$, giving $\beta = v_2(3c+1) - 1 = v_2(3^{\alpha+1}m - 1) - 2$. □

3.3 Modular characterisation of β

Proposition 3.4 (Modular characterisation of $\beta = 1$)**.** *With $a+1 = 2^\alpha m$ (m odd) and $c = (3^\alpha m - 1)/2$, the following are equivalent:*

(i) $\beta = 1$*;*

(ii) $c \equiv 1 \pmod 8$*;*

(iii) $3^\alpha m \equiv 3 \pmod{16}$.

Proof. By Lemma 3.3, $\beta = 1$ iff $v_2(3^{\alpha+1}m - 1) = 3$, i.e. $3^{\alpha+1}m - 1 \equiv 8 \pmod{16}$.

(i) ⇔ (ii). Since $3c + 1 = (3^{\alpha+1}m - 1)/2$, we have $\beta = 1$ iff $v_2(3c+1) = 2$, i.e. $3c + 1 \equiv 4 \pmod 8$, i.e. $3c \equiv 3 \pmod 8$, i.e. $c \equiv 1 \pmod 8$.

(ii) ⇔ (iii). $c \equiv 1 \pmod 8$ iff $3^\alpha m - 1 \equiv 2 \pmod{16}$ iff $3^\alpha m \equiv 3 \pmod{16}$. □

Corollary 3.5 (Equivalence with orbit structure)**.** *Let $a+1 = 2^\alpha m$, $c = \Phi_{\mathrm{St}(\alpha,1)}(a)$, $b = \Phi_{\mathrm{St}(1,\beta)}(c)$. The following are equivalent:*

(A) $c \equiv 1 \pmod 8$ *(the next Steiner word after* $\mathrm{St}(\alpha,1)$ *is* $\mathrm{St}(1,1)$*);*

(B) $3^\alpha m \equiv 3 \pmod{16}$*;*

(C) $b = \Phi_{\mathrm{St}(\alpha+1,3)}(2a+1)$.

Proof. (A) ⇔ (B) is Proposition 3.4. (A) ⇔ (C): $\beta = 1$ iff the identity (1) gives $b = \Phi_{\mathrm{St}(\alpha+1,3)}(2a+1)$, and condition (A) is exactly the orbit-level statement that $\mathrm{St}(1,1)$ is the next word. □

Remark 3.6. The general case $\beta \geq 0$ follows the same pattern: β is the unique value satisfying $v_2(3^{\alpha+1}m - 1) = \beta + 2$, i.e. $3^{\alpha+1}m - 1 \equiv 2^{\beta+2} \pmod{2^{\beta+3}}$, and the identity (1) then expresses b in terms of $2a+1$ via $\mathrm{St}(\alpha+1, \beta+2)$.

4 Related Work

This paper depends entirely on the structural polynomial machinery of [2]. The $2a+1$ sibling identity in Collatz orbit analysis arises naturally in the Steiner word decomposition of [1]; the present uniqueness theorem provides the algebraic explanation for why it is special to the Collatz basis. The $2a+1$ identity is a structural feature of the Collatz sibling relation, not a generic property of (g, h) systems.

References

[1] J. Seymour, *Steiner Word Decomposition and Branch Length Distribution*, preprint, 2026. https://wildducktheories.github.io/collatz/papers/67-branch-length-distribution/67-branch-length-distribution.pdf

[2] J. Seymour, *The Structural Polynomial of a Parity Sequence and Its Applications*, preprint, 2026. https://wildducktheories.github.io/collatz/papers/71-structural-polynomial/71-structural-polynomial.pdf